

## FORT VERMILION, AB, Canada

WMO: 710240

Lat: 58.3822N

Lon: 116.0403W

Elev: 948

StdP: 14.20

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -31.6      | -26.2 | -38.6                         | 0.6 | -31.4 | -32.5 | 0.9 | -26.1 | 14.5                  | 18.1 | 12.3 | 10.1 | 1.9                   | 300  | 0.594 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.3                   | 83.9            | 64.4 | 80.9 | 62.8 | 77.9 | 61.2 | 67.4                | 79.0 | 65.6 | 76.6 | 63.8 | 73.9 | 4.9                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 63.6                            | 91.0 | 71.1 | 61.5 | 84.6 | 69.1 | 59.7 | 79.1 | 67.5 | 32.4          | 78.6 | 30.9 | 76.9 | 29.5 | 74.2 | 76.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|
| 1%                |      | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |
| Min               |      | Max  |                            | Min   | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 12.3              | 10.8 | 9.6  | DB                         | -39.8 | 90.3 | 6.9                | 3.0                                                | -44.8     | 92.5 | -48.9      | 94.2 | -52.7      | 95.9 | -57.8      | 98.2 |
|                   |      |      | WB                         | -39.3 | 70.3 | 6.8                | 3.2                                                | -44.2     | 72.6 | -48.2      | 74.5 | -52.0      | 76.3 | -56.9      | 78.6 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 33.4   | -0.3  | 4.5   | 17.2  | 37.3  | 51.9  | 60.8  | 64.7  | 60.6  | 50.2  | 35.9  | 14.6  | 1.0   |
|                                                                                  | DBStd       | 25.96  | 14.09 | 13.13 | 14.72 | 10.43 | 9.06  | 6.18  | 5.18  | 6.24  | 7.79  | 8.54  | 12.09 | 13.34 |
|                                                                                  | HDD50       | 7452   | 1558  | 1274  | 1015  | 394   | 84    | 6     | 0     | 4     | 90    | 446   | 1063  | 1518  |
|                                                                                  | HDD65       | 11669  | 2023  | 1694  | 1480  | 832   | 412   | 151   | 69    | 163   | 445   | 904   | 1513  | 1983  |
|                                                                                  | CDD50       | 1375   | 0     | 0     | 0     | 12    | 143   | 329   | 455   | 332   | 97    | 7     | 0     | 0     |
|                                                                                  | CDD65       | 117    | 0     | 0     | 0     | 0     | 6     | 24    | 59    | 26    | 2     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 1672   | 0     | 0     | 0     | 8     | 155   | 370   | 735   | 356   | 46    | 2     | 0     | 0     |
|                                                                                  | CDH80       | 359    | 0     | 0     | 0     | 0     | 26    | 65    | 190   | 73    | 5     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 4.1    | 3.6   | 3.8   | 4.5   | 5.1   | 5.0   | 4.0   | 3.7   | 3.7   | 3.9   | 4.4   | 3.9   | 3.4   |
| Precipitation                                                                    | PrecAvg     | 16.1   | 0.9   | 0.8   | 0.9   | 0.8   | 1.5   | 1.9   | 2.7   | 2.3   | 1.5   | 1.1   | 0.9   | 0.9   |
|                                                                                  | PrecMax     | 22.6   | 2.5   | 1.5   | 2.3   | 2.8   | 3.8   | 5.2   | 7.0   | 4.5   | 3.8   | 3.3   | 1.5   | 1.8   |
|                                                                                  | PrecMin     | 11.6   | 0.2   | 0.1   | 0.1   | 0.0   | 0.2   | 0.3   | 0.7   | 0.6   | 0.4   | 0.1   | 0.0   | 0.4   |
|                                                                                  | PrecStd     | 3.2    | 0.5   | 0.4   | 0.6   | 0.7   | 0.9   | 1.3   | 1.5   | 1.2   | 0.9   | 0.8   | 0.5   | 0.4   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 37.9  | 41.2  | 53.3  | 72.4  | 83.6  | 85.9  | 89.5  | 86.3  | 79.8  | 68.1  | 45.6  | 36.4  |
|                                                                                  |             | MCWB   | 34.3  | 35.0  | 41.6  | 52.0  | 58.3  | 64.4  | 67.5  | 66.0  | 63.0  | 54.2  | 38.2  | 32.6  |
|                                                                                  | 2%          | DB     | 29.4  | 34.4  | 47.3  | 63.3  | 78.2  | 81.5  | 84.7  | 81.6  | 73.1  | 58.5  | 39.2  | 28.8  |
|                                                                                  |             | MCWB   | 27.8  | 30.3  | 37.7  | 47.5  | 55.6  | 61.1  | 65.9  | 64.4  | 58.1  | 47.8  | 34.8  | 27.4  |
|                                                                                  | 5%          | DB     | 24.0  | 27.9  | 43.1  | 58.2  | 74.1  | 78.4  | 81.5  | 77.9  | 68.3  | 53.2  | 34.3  | 23.9  |
|                                                                                  |             | MCWB   | 23.2  | 25.1  | 35.2  | 44.2  | 53.8  | 60.3  | 64.9  | 62.4  | 55.8  | 44.7  | 32.2  | 23.3  |
|                                                                                  | 10%         | DB     | 18.6  | 23.1  | 38.5  | 53.9  | 69.4  | 75.0  | 78.0  | 74.4  | 64.0  | 48.6  | 30.3  | 19.6  |
|                                                                                  |             | MCWB   | 18.1  | 21.3  | 32.6  | 42.1  | 51.5  | 59.1  | 63.1  | 60.6  | 53.4  | 42.4  | 29.2  | 19.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 34.2  | 35.8  | 41.6  | 52.4  | 60.1  | 67.7  | 71.3  | 69.2  | 64.5  | 55.3  | 39.0  | 33.4  |
|                                                                                  |             | MCDB   | 37.4  | 41.2  | 52.2  | 70.7  | 79.2  | 78.6  | 82.5  | 81.3  | 77.3  | 66.8  | 43.5  | 35.5  |
|                                                                                  | 2%          | WB     | 28.0  | 30.0  | 38.4  | 48.2  | 57.5  | 64.9  | 68.0  | 66.6  | 60.4  | 49.0  | 35.3  | 27.4  |
|                                                                                  |             | MCDB   | 29.3  | 33.7  | 46.3  | 62.0  | 74.7  | 75.9  | 80.4  | 76.4  | 69.7  | 55.9  | 38.5  | 28.9  |
|                                                                                  | 5%          | WB     | 23.1  | 25.2  | 35.6  | 45.0  | 54.8  | 62.6  | 66.3  | 64.4  | 57.6  | 45.4  | 32.2  | 23.4  |
|                                                                                  |             | MCDB   | 24.2  | 27.6  | 42.3  | 56.3  | 70.6  | 73.2  | 78.3  | 74.4  | 65.2  | 51.9  | 34.0  | 24.0  |
|                                                                                  | 10%         | WB     | 18.1  | 21.2  | 32.7  | 42.5  | 52.5  | 60.8  | 64.7  | 62.4  | 55.0  | 42.7  | 29.2  | 18.9  |
|                                                                                  |             | MCDB   | 18.6  | 23.1  | 38.1  | 52.9  | 66.9  | 71.1  | 75.4  | 71.7  | 61.8  | 47.8  | 30.4  | 19.6  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.3  | 19.9  | 22.8  | 22.0  | 24.0  | 22.2  | 22.3  | 22.4  | 21.2  | 15.3  | 13.3  | 14.3  |
|                                                                                  |             | MCDBR  | 21.8  | 25.3  | 26.7  | 28.6  | 31.3  | 28.5  | 28.7  | 28.5  | 28.2  | 24.5  | 16.6  | 17.0  |
|                                                                                  |             | MCWBR  | 20.1  | 21.3  | 18.1  | 15.3  | 13.4  | 11.8  | 12.5  | 12.8  | 14.5  | 14.7  | 13.2  | 15.7  |
|                                                                                  | 5% WB       | MCDBR  | 21.6  | 25.0  | 25.7  | 26.9  | 27.7  | 23.6  | 25.7  | 25.7  | 25.4  | 22.8  | 15.9  | 16.8  |
|                                                                                  |             | MCWBR  | 19.9  | 21.1  | 17.7  | 14.9  | 12.5  | 10.9  | 12.2  | 12.8  | 13.8  | 14.0  | 13.1  | 15.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.222  | 0.246 | 0.272 | 0.331 | 0.342 | 0.361 | 0.369 | 0.348 | 0.311 | 0.283 | 0.233 | 0.203 |       |
|                                                                                  | taud        | 2.235  | 2.313 | 2.301 | 2.333 | 2.426 | 2.401 | 2.389 | 2.447 | 2.546 | 2.559 | 2.339 | 2.122 |       |
|                                                                                  | Ebn at Noon | 209    | 259   | 281   | 279   | 283   | 278   | 272   | 271   | 267   | 245   | 205   | 172   |       |
|                                                                                  | Edh at Noon | 14     | 21    | 29    | 33    | 33    | 34    | 34    | 30    | 23    | 17    | 12    | 11    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 180    | 478   | 1008  | 1549  | 1813  | 1889  | 1788  | 1448  | 981   | 481   | 191   | 108   |       |
|                                                                                  | RadStd      | 13     | 32    | 62    | 74    | 136   | 142   | 81    | 91    | 71    | 59    | 21    | 8     |       |

## Historical Trends

|                        |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (5 neighbors) |  | N/A   | +2.86   | +3.83  | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page